

Data Preparation II (continued)

Building a Panel Dataset

Dr. Fidel González

Sam Houston State University

Spring 2026

Agenda (and where we are)

This lecture corresponds to **Data Preparation II (continued)**.

Today we will:

- Build a real panel dataset from raw economic data
- Understand why panels are *constructed*, not downloaded
- Diagnose duplicates, gaps, and coverage problems
- Document cleaned data for future analysis

Big picture: Good analysis starts with a defensible data structure.

How Lecture 5 builds on Lecture 4

Last time, we learned:

- How merges work and how they fail
- Why keys determine whether results are correct
- Why reshaping enables analysis

Today's shift:

- From combining datasets *once*
- To building a dataset that works *over time*

Key idea: Panels make mistakes persistent if you don't catch them early.

What is a panel dataset?

A panel dataset combines:

- Many units (e.g., counties)
- Many time periods (e.g., years)

Each row represents:

Unit $\times$ Time

Examples:

- County–year unemployment
- Firm–quarter profits
- Individual–wave survey responses

Keys in panel data

In a panel, the key must identify:

- **Who** the observation is
- **When** it occurs

Examples:

- County panel: `state_fips + county_fips + year`
- Firm panel: `firm_id + quarter`

Rule:

If you cannot clearly state the panel key, you do not have a panel.

Why building panels is hard

Common real-world problems:

- Units appear or disappear over time
- Variable definitions change
- Missing years are not random
- Historical data get revised

Economist's job: Decide what to keep, what to drop, and what to document.

Case study: County-level unemployment

We will build a panel using:

- County-level unemployment data
- Multiple years
- A single outcome variable

Why this case?

- Policy relevance
- Familiar concept
- Messy enough to matter

Step 1: Start with one year

Before building a panel:

- Load a single year
- Verify key uniqueness
- Check missing values

Principle: If one year is broken, ten years will be worse.

Step 2: Stack years vertically

Panels grow **down**, not across.

You will:

- Load multiple years
- Stack them (append / concatenate)
- Create a year variable explicitly

This is not a merge problem — yet.

Step 3: Diagnose the panel

After stacking, always check:

- Are keys unique?
- Do all units appear every year?
- Are there gaps in time?

This tells you whether the panel is:

- Balanced
- Unbalanced

Balanced vs unbalanced panels

Balanced panel:

- Every unit observed every period

Unbalanced panel:

- Some units missing in some periods

Most applied work uses **unbalanced panels**. What matters is transparency.

Panel data QA: professional habits

Always check:

- Duplicate unit–time pairs
- Coverage changes over time
- Systematic missingness

Rule:

If you don't diagnose your panel, someone else will.

Documenting your panel

A clean panel should include:

- Variable definitions
- Units of measurement
- Data sources
- Known limitations

This can be:

- A markdown cell
- A README
- A mini data dictionary

Key takeaways

- Panels are constructed, not downloaded
- Keys define truth in panel data
- Unbalanced panels are normal
- QA and documentation are part of analysis

Structure first. Model second.

Questions?